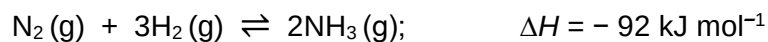


Section A

Answer **all** the questions in this section in the spaces provided.

- 1 Nitrogen and hydrogen react to form ammonia in the Haber process.



Le Chatelier's principle predicts that the highest equilibrium yield of ammonia in this process should occur at a high pressure and at a low temperature; however, these conditions are not actually used in the industry.

- (a) (i) State the conditions used in the Haber process.

Pressure :

Temperature :

- (ii) The rate of the reaction could be increased by passing the reactants through a tower packed with lumps of iron. State and explain the function of iron.

.....

.....

.....

.....

[5]

- (b) During the synthesis of ammonia, the following equilibrium concentrations were obtained.

molecule	Equilibrium concentration / mol dm ⁻³
N ₂	1.383
H ₂	1.145
NH ₃	0.236

Calculate the value of K_c and state its units for this reaction.

[2]

- (c) Besides the above factors, state and explain one other way that may bring about a greater yield of NH_3 (g).

.....

[2]

[Total: 9]

- 2 (a) (i) Define *enthalpy change of combustion*.

.....

- (ii) Write the chemical equation, including state symbols, that represents the enthalpy change of combustion of butane gas.

.....

- (iii) Using bond energy values from the *Data Booklet*, calculate the enthalpy change of combustion of butane gas.

[4]

- (b) When butane reacts with chlorine gas under certain conditions, various substituted chloro products are formed.

- (i) State the type of reaction that occurs between chlorine gas and butane gas.

.....

- (ii) State the condition required for the above reaction to occur.

.....[2]

(c) Mono-substituted chlorobutane undergoes elimination to form an alkene, C_4H_8 . One of the products of this elimination reaction exists as geometrical isomers.

(i) State the reagents and conditions needed for this elimination reaction to occur.

.....

(ii) Name and draw the **displayed** formulae of the geometrical isomers of C_4H_8 .

[3]

(d) Explain why benzene undergoes substitution reactions whereas butene undergoes addition reactions.

.....

.....

.....

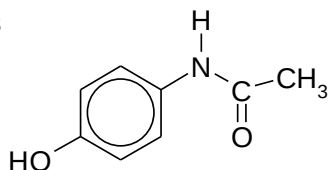
.....

.....[3]

[Total: 12]

- 3 Paracetamol (common brand name: Panadol) is a drug that can be purchased over-the-counter at most pharmacies. The drug is used to relieve pain and reduce fevers. Paracetamol is broken down by the liver. Thus too much paracetamol in the body may lead liver failure.

The structure of paracetamol is



Pure paracetamol is a white crystalline solid which melts at 169 °C. Its solubility in water is 1.43 g/100 cm³, and 14 g/100 cm³ in ethanol.

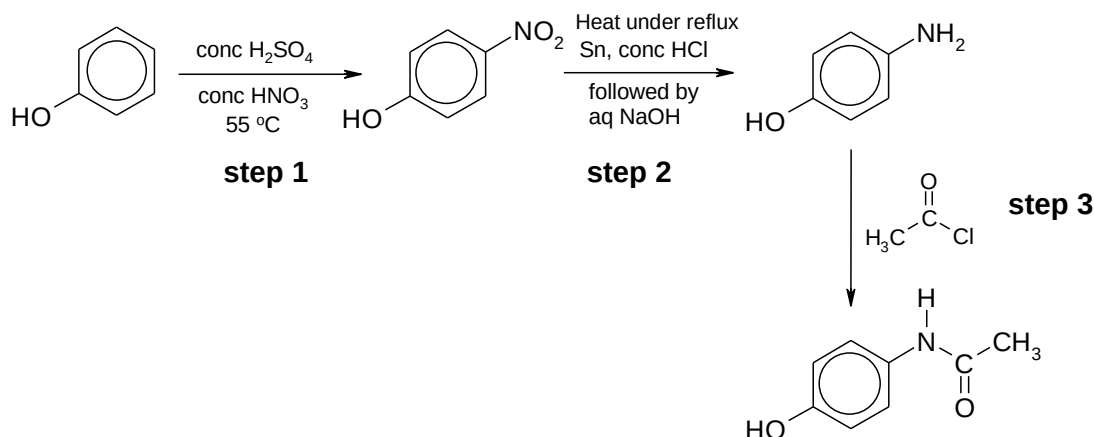
- (a) Suggest a reason for the limited solubility of paracetamol in water.

.....

[2]

Preparation of Paracetamol

Paracetamol can be made in three steps from phenol.



- (b) (i) State the type of reaction in **steps 1** and **2**.

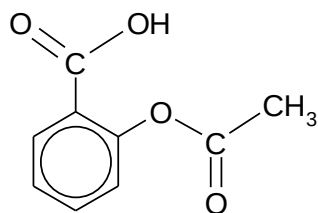
Step 1:

Step 2:

- (ii) Write a balanced equation for **step 2**. The symbol [O] and [H] are acceptable.

.....[3]

- (c) Another painkiller, aspirin, is as widely used as paracetamol. The structure of aspirin is as follows:



Each of these two drugs has a group which is easily hydrolysed in the body. For example, aspirin can be digested by enzyme in the intestine.

- (i) Name the functional group that is hydrolysed in aspirin.

.....

- (ii) There is a **similar** group in paracetamol which can be hydrolysed. In the spaces below, draw the structural formulae of the two hydrolysis products for each drug.

paracetamol	aspirin

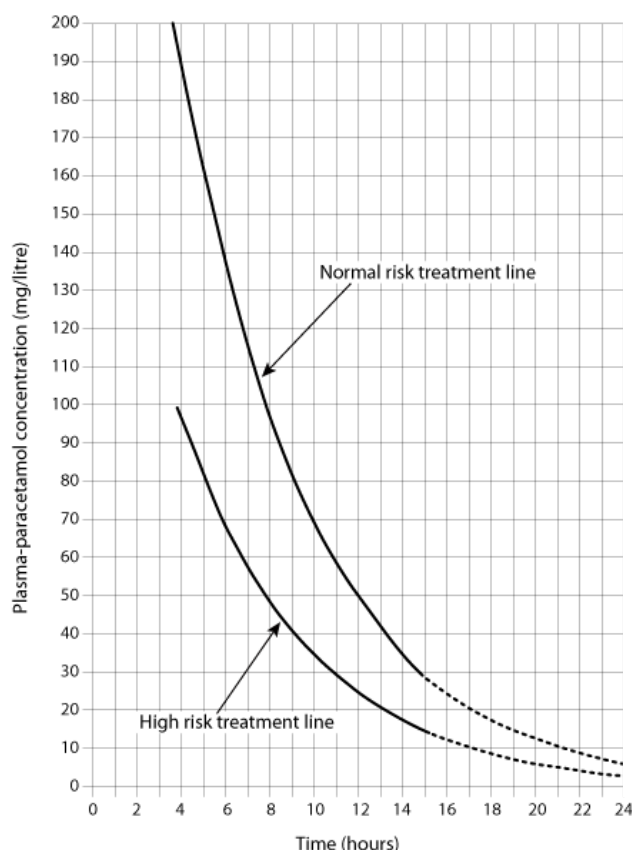
[5]

The daily recommended dose of paracetamol for a healthy adult is 500 mg –1000 mg every 4 hours. An overdose occur if a person took more than 4 g of paracetamol in any 24-hour period.

Paracetamol is perceived to be a safe drug. A hidden risk in paracetamol overdose lies in the fact that there are many **other** common products **containing** paracetamol available over-the-counter. This is worsened by the common practice of people who would work on a “top-up” dosage when they do not get better.

Paracetamol overdose, even in small amounts can cause fatal liver damage. A blood sample is tested for blood-paracetamol (**plasma-paracetamol**) level at least 4 hours after ingestion to determine the likelihood of liver injury and the need for antidote (N-acetylcysteine) treatment. Results of this blood test are plotted on the **Paracetamol Treatment Graph** as shown in **Figure 1**.

Figure 1



There are two treatment lines in the **Paracetamol Treatment Graph**.

- The normal risk treatment line for healthy adults;
- The high risk treatment line for patients with a liver condition, malnourished, or takes chronic alcohol in excess.

Patient should be treated if their plasma-paracetamol concentration is **above** their appropriate treatment line.

(d) A person has a blood test after 8 hours of paracetamol ingestion and it was found to be 80 mg/litre.

(i) Using the Paracetamol Treatment Graph in **Figure 1**, plot the result of the person's blood test using 'X' as a symbol.

(ii) Deduce if the person needs antidote treatment, include any assumptions made.

.....

.....[2]

- (ii) Discuss why paracetamol overdose is common.

.....
[2]

(e) A case study of paracetamol overdose

A young man was admitted to a hospital with the claim of paracetamol overdose. The following information was available:

- He has no other established medical condition and he does not consume alcohol.
- He had taken twelve 500 mg(milligrams) Panadol tablets in the past 12 hours for a severe and persistent headache.
- Plasma-paracetamol test was done 1 hour after admission.
- An adult human male has about 5 litres of blood in his body.
- By detoxification action of the liver, paracetamol has a half-life of 27 hours in the human body.

- (i) Based on the recommended dose, explain whether the young man's claim of paracetamol overdose is valid.

.....
[1]

- (ii) Why is **(e)(i) alone** insufficient to confirm on a case of overdose?

.....
[2]

- (iii) By performing appropriate calculations, suggest and explain whether this patient should be given the antidote treatment.

[2]
 [Total: 19]

Section B

Answer **two** questions from this section on separate answer paper.

- 4 About 90 % of the ozone in the Earth's atmosphere is found in the region called the stratosphere. The stratosphere is approximately 30 km above the Earth's surface. Ozone is being made continuously by the following reaction 1:

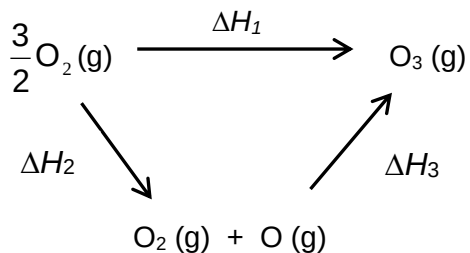


The free oxygen atoms arise from the splitting of oxygen molecules by ultraviolet light from the Sun, according to reaction 2:

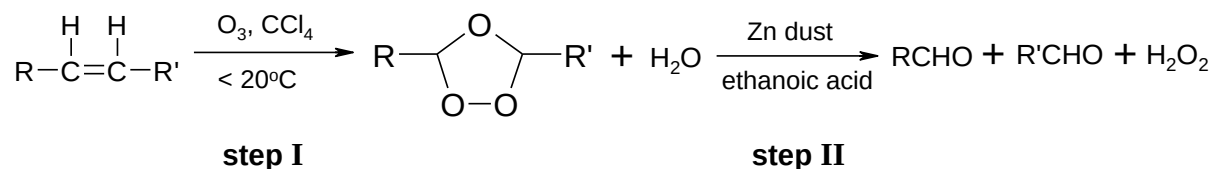


- (a) (i) In reaction 1, the order of the reaction with respect to $[\text{O}_2]$ is one, **sketch** a graph of $[\text{O}_2]$ against time and use it to show how the order of reaction with respect to O_2 is derived from this sketch. Label your sketch fully.
- (ii) Experimental results show that when the $[\text{O}]$ is doubled, while keeping $[\text{O}_2]$ constant, the initial rate of reaction is also doubled. Write the rate equation for the formation of ozone.
- (iii) Calculate the rate constant, including units, at which ozone forms, given the following values:
 Rate = $1.52 \times 10^{-22} \text{ mol dm}^{-3} \text{ s}^{-1}$; $[\text{O}] = 3 \times 10^{-14} \text{ mol dm}^{-3}$; $[\text{O}_2] = 1.3 \times 10^{-4} \text{ mol dm}^{-3}$.
[5]
- (b) The temperature of the stratosphere is -50°C .
- (i) Explain qualitatively how the rate of ozone formation at sea level would compare with that in the stratosphere.
- (ii) Practically no ozone is formed at sea level. Suggest a reason for this. [3]
- (c) The ozone molecule contains a covalent bond known as a dative bond.
- (i) Draw a dot-and-cross diagram of the ozone molecule, O_3 .
- (ii) Suggest a value for the bond angle in the ozone molecule.
- (iii) The boiling point of oxygen gas is -183°C . Suggest and explain whether the boiling point of ozone will be higher or lower than oxygen gas. [4]

- (d) The overall reaction for the formation of ozone is given as $3\text{O}_2(\text{g}) \rightarrow 2\text{O}_3(\text{g})$. An energy cycle is given below.



- (i) What does the term ΔH_1 in the above energy cycle represent?
- (ii) Using the *Data Booklet*, suggest a value for ΔH_2 .
- (iii) Given that ΔH_1 is $+142.7 \text{ kJ mol}^{-1}$, calculate ΔH_3 . [4]
- (e) Ozone can be used in organic chemistry to convert an alkene to aldehydes in the reaction scheme given below.



- (i) State the type of reaction that occurs in step II.
- (ii) When ozone is added to but-2-ene, ethanal is formed. Suggest a **three-step** reaction scheme, **other than the one given above**, to convert but-2-ene to ethanal, stating clearly the reagents and conditions used in each of the three steps and the structure of the reaction intermediate.

[4]

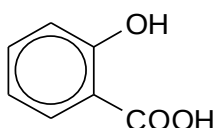
[Total: 20]

5 (a) The oxides Na_2O , Al_2O_3 , SiO_2 and P_4O_{10} differ considerably in their physical and chemical properties:

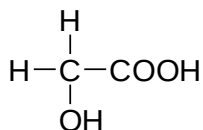
- (i) In terms of structure and bonding, explain why P_4O_{10} has a low melting point whereas the other oxides have high melting points;
- (ii) Na_2O and P_4O_{10} dissolve in water whereas the other oxides do not. Describe the reactions of Na_2O and P_4O_{10} with water. Write equations where appropriate.
- (iii) Using the above oxides as examples, describe the acid-base nature of the oxides by their reaction with acid or alkali. Write equations where appropriate.

[10]

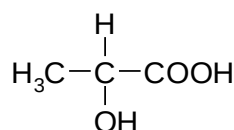
(b) Salicylic acid, glycolic acid and lactic acid are three organic acids commonly used in cosmetic preparations.



Salicylic acid



Glycolic acid



Lactic acid

Suggest a simple chemical test that can be used to distinguish glycolic acid from lactic acid.

State the reagents and conditions, give the expected observations and write balanced equations for the reactions that occur. [4]

(c) A monocarboxylic acid **A** is found to be present in goat's milk. **A** has the following composition by mass:

C, 62.1 %; H, 10.3 %; O, 27.6 %.

Neutralisation of a sample of 0.10 g of **A** requires 8.6 cm³ of 0.10 mol dm⁻³ sodium hydroxide.

A can be synthesised in the laboratory from 1-bromopentane by a 2-stage reaction sequence:



- (i) Calculate the empirical formula and the molecular formula of **A**.
- (ii) Suggest the reagents and conditions necessary for each step of the synthesis.
- (iii) Predict from this synthesis the structural formulae of **A** and **B**.

[6]

[Total: 20]

6 This question is about acids. Johannes Brønsted and Martin Lowry independently came to similar conclusions about acid-base behaviour early in the 20th century.

(a) (i) Describe what is meant by a brønsted acid and a brønsted base.

(ii) Write an equation for the reaction between hydrogen chloride, HCl , and water. Indicate which species are the acid, the base and their conjugate pairs in the reaction. [3]

(b) A 20.0 cm^3 sample of $0.200 \text{ mol dm}^{-3}$ CH_3COOH (aq) was titrated with $0.100 \text{ mol dm}^{-3}$ NaOH .

(i) Calculate the pH of $0.200 \text{ mol dm}^{-3}$ CH_3COOH (aq) given $K_a = 1.75 \times 10^{-5} \text{ mol dm}^{-3}$.

(ii) Explain whether you would consider CH_3COOH a weak or strong acid.

(iii) Sketch a graph to illustrate how the pH changes during this titration. Mark clearly on your graph,

1. where the end point occurs,
2. where the mixture is being buffered at maximum capacity.

(iv) Explain with the help of equations how the buffer system works. [7]

(c) Carboxylic acids and their derivatives can be found in many common products. Ester flavouring agents are synthesised industrially to have flavours very close to the target flavour to make ice-cream, soft drinks or lollipops taste natural.

Ester **A** (Mr, 116) is used in synthetic fruit flavours. When **A** is subjected to hydrolysis, a monoprotic acid **B** (Mr, 60) and an alcohol **C** are produced.

Alcohol **C** reacts with alkaline aqueous iodine giving a precipitate **D**. **C** is oxidised to ketone **E**. This ketone also reacts with alkaline aqueous iodine, giving the same precipitate **D** and a solution containing a sodium salt **F**.

(i) Deduce the structure of alcohol **C**.

(ii) State the type of reaction between **C** and **E** with alkaline aqueous iodine, identifying **D** and **F**.

(iii) Suggest the structural formulae of **E**.

(iv) Hence, identify the original ester **A** and write a balanced equation for its hydrolysis.

[10]

[Total: 20]

END OF PAPER